UDEE Practice Test 06: Answer Key

30 questions · Section A Figural, Section B Observation (interference), Section C Logic-Based Reasoning · Printed separately from the test sheet

Review rules. Read the explanation for every question you missed AND every question you guessed correctly. Understanding why the trap options are wrong matters more than getting the right answer once. Flag any question where the explanation surprises you; that gap is what the real UDEE will exploit.

Answer Grid

1. D	2. D	3. B	4. C	5. E
6. E	7. C	8. A	9. D	10. B
11. B	12. C	13. B	14. C	15. B
16. C	17. B	18. C	19. A	20. B
21. C	22. A	23. B	24. B	25. C
26. B	27. C	28. A	29. C	30. C

Section A: Figural Reasoning

Q1: D. Rotated 180 degrees

A 180 degree rotation flips the figure both horizontally and vertically about its center. The original L has a vertical bar on the LEFT with a horizontal foot at the BOTTOM extending right. After 180 degree rotation: vertical bar on the RIGHT with a horizontal bar at the TOP extending left. Option A is unchanged (no rotation). Option B is flipped top-to-bottom only (horizontal flip across a horizontal axis). Option C is 90 degrees clockwise (horizontal bar on top with a narrow vertical drop on the left). Option E is mirrored left-to-right only.

Q2: D. 8 triangles

A square with both diagonals produces exactly 8 triangles. Count them by size. There are 4 small triangles inside the square where the diagonals divide it (top, bottom, left, right). Then there are 4 larger triangles, each formed by taking half the square along one diagonal: top-half, bottom-half, left-half, right-half. Total `4 + 4 = 8`. Option A (4) counts only the small interior triangles. Option B (6) and C (7) are common partial counts.

Q3: B. Black pentagon

Two rules operate independently. Rows change shape: row 1 has triangles, row 2 has squares, row 3 has pentagons. Columns change fill color: column 1 is white, column 2 is gray, column 3 is black. The missing cell at row 3, column 3 therefore combines pentagon (row 3) with black (column 3). Option A keeps the shape right but wrong color. Option C swaps to hexagon. Option D reverses the color. Option E takes the wrong shape from row 2.

Q4: C. 5

On a standard die, opposite faces sum to 7. Top = 2, so bottom = `7 - 2 = 5`. The front and right faces are irrelevant for computing the bottom. Option B (4) would be correct only if the top were 3. Option E ignores the rule that makes the bottom fully determined from the top alone.

Q5: E. 720

The sequence is factorials: `1! = 1, 2! = 2, 3! = 6, 4! = 24, 5! = 120, 6! = 720`. Each term equals the previous term multiplied by the next integer: `1 × 2 = 2`, `2 × 3 = 6`, `6 × 4 = 24`, `24 × 5 = 120`, `120 × 6 = 720`. Option A (240) is `120 × 2`, a common misread. Option D (600) is `120 × 5`, same multiplier as the previous step.

Q6: E. The J-pentomino

Figures A through D are all the L-pentomino (four cells vertical with one cell offset at the base) in different rotations. Figure E is the J-pentomino, the MIRROR IMAGE of the L-pentomino. Reflections are not allowed by rotation alone, so E cannot be produced from A through D. This is the same logic as Test 05 Q6 with the S-tetromino and Z-tetromino: mirror-image pentominoes look similar but are chirally distinct.

Q7: C. 8 holes

Each fold in half doubles the layer count: fold 1 = 2, fold 2 = 4, fold 3 = 8. One punch through 8 layers creates 8 holes when the paper is unfolded. Option E (16) would require four folds, not three.

Q8: A. Only the small central region is shaded

The triple-overlap region is where all three circles intersect at once. In a symmetric three-circle arrangement it is a small central roughly-triangular region. Option B shades all three full circles (the union). Option C shades the entire overlap of only two circles (a pairwise lens). Option D shades the full union again. The question explicitly asks for the region inside ALL three circles simultaneously, which is the smallest region A depicts.

Q9: D. West

Rotating 90 degrees clockwise from North cycles through: N → E → S → W → N. So four steps return to North. Seven steps: steps 1-4 = E, S, W, N; steps 5-7 = E, S, W. After step 7, the pointer faces West. Formally: `(7 × 90) mod 360 = 270 degrees clockwise from N = West`.

Q10: B. Triangle inverted with dot near bottom apex

Reflection across a horizontal axis (the water surface) flips the figure top-to-bottom. The original triangle points UP with a dot near the top. The reflection points DOWN with the dot near the bottom apex. Option A inverts the triangle but leaves the dot near the top, which is inconsistent with a simple mirror. Option C shows no reflection. Options D and E are rotations, not reflections.

Section B: Observation with Interference

Q11: B. Embassy of Tevera

The dark blue sign over the embassy building read "EMBASSY OF TEVERA" in gold letters, and the flag carried the words "REPUBLIC OF TEVERA." Option A and D are plausible-sounding country names chosen to catch candidates who retained only the "T" initial. Option C (Savonia) is a distractor with no overlap.

Q12: C. 14:23

The digital clock mounted on the embassy facade displayed `14:23` in red LED. Scene 12 later showed 8:15 PM on an analog clock; do not let that later detail overwrite this one.

Q13: B. DPL-7419

The middle vehicle (the limousine carrying the VIP, marked with a gold border in the scene) carried plate `DPL-7419`. Option A transposes digits to 4719, a classic number-swap trap. Options C and D look like domestic plates and should be easy to reject for a diplomatic vehicle (DPL implies Diplomatic).

Q14: C. 3

Three agents in dark suits with visible earpieces were positioned around the motorcade: one at the limousine door (left), one behind the VIP (middle), and one at the rear of the limousine (right). The K-9 handler far-left and the press pool right-side are separate roles, not protective detail agents.

Q15: B. A K-9 unit with a handler

Far left of the scene, a uniformed K-9 handler stood with a tan working dog on a leash. No robot, SWAT team, or mounted unit appeared. This question tests whether you correctly identify the support unit even though it was off to one side of the main motorcade action.

Q16: C. 8:15 PM

The analog wall clock on the left wall of the dining room showed hands pointing to 8:15, with a small white label reading "8:15 PM" underneath. Scene 11's 14:23 is the interference trap; always bind the time to the correct scene.

Q17: B. A dark blue suit with a gold lapel pin

The VIP at the head of the table wore a dark blue suit jacket with a visible small gold pin on the lapel. Option A (tuxedo and bow tie) is the ceremonial distractor; options D and E invent details that were not present. This question rewards candidates who retained both the color AND the pin as paired details.

Q18: C. 4 guests

> Four guests sat at the table: two on each side. The VIP at the head plus four guests makes five people seated, with three additional agents (one behind the VIP, one at each side wall) standing. The question explicitly excludes the VIP and the agents, so the count is 4.

Q19: A. A silver tray with three glasses

> The server approaching from the right carried a silver tray holding three small glasses (drawn as gold-filled rectangles on the tray, representing drinks). Options B through E describe plausible food-service alternatives that were not shown.

Q20: B. Scene 12 only, showing 8:15

> Scene 11 had a DIGITAL clock (14:23 in red LED). Scene 12 had an ANALOG clock (hands at 8:15). Only Scene 12 fits the "analog clock" criterion. Option A misreads Scene 11's digital as analog. Option C claims both had analog clocks, which is wrong. Option E is a trap for candidates who mistake digital format for analog.

Section C: Logic-Based Reasoning

Q21: C. The crew was ready AND the runway was clear

> The rule is a biconditional: `on_time ⇔ (crew_ready ∧ runway_clear)`. Given the flight DID depart on time (the positive case, not the negative), both sides of the biconditional must match. Since on_time is true, the right side must be true: `crew_ready ∧ runway_clear`. Both conditions confirmed. Option A drops one condition; B drops the other. Option D is too weak (says "at least one" when the conjunction requires BOTH). Option E contradicts the biconditional. Test 05 Q21 worked in the other direction (vault did NOT open, derive what failed); test 06 Q21 works forward (flight DID depart, derive what held).

Q22: A. C, B, E, F, D, A

> Verify each constraint against C, B, E, F, D, A:

(1) B is in position 2: position 2 is B. OK.

(2) D is in position 5: position 5 is D. OK.

(3) E and F adjacent: E at 3, F at 4, consecutive. OK.

(4) C before D: C at 1, D at 5. OK.

(5) A not in position 1: A at 6. OK.

Option B places A at position 1, violating (5), and C at position 6 (after D), violating (4). Option C places E at position 5 where D must be, violating (2). Option D places B at position 1, violating (1). Option E places D at position 6 instead of 5, violating (2).

Q23: B. 7

Pigeonhole: with 45 calls across 7 operators, the best-distributed arrangement has some operators handling 7 and others 6. Check: `45 / 7 = 6.43`. So `⌈45/7⌉ = 7`. Construction: 7+7+7+6+6+6+6 = 45 with max = 7. Option A (6) fails because 7 operators of 6 = 42, short by 3. **Trap:** rounding DOWN 6.43 to 6 is the most common mistake.

Q24: B. Adams and Carr

Test all three possible truth-pairs under "exactly two truthful":

Adams + Brooks: Adams says Brooks is lying, but Brooks is supposed to be truthful. Direct contradiction.

Adams + Carr: Adams says Brooks lies (consistent). Carr says Adams is truthful (consistent). Brooks lies, so his statement "Carr is lying" is false, so Carr is truthful. All three conditions hold. OK.

Brooks + Carr: Brooks says Carr is lying, but Carr is supposed to be truthful. Contradiction.

Only {Adams, Carr} works.

Q25: C. Both Gate A and Gate B are closed

The rule is `(A_open ∨ B_open) → alarm`. Apply modus tollens on NOT alarm: `¬alarm → ¬(A_open ∨ B_open) = ¬A_open ∧ ¬B_open`. Both gates must be closed. This is the De Morgan contrapositive of a disjunction. Option D says "at least one closed," which is too weak; the premise forces BOTH closed. Options A and B each pick only one gate.

Q26: B. South is NOT on red alert

Exclusive OR: exactly one of {North, South} is on red alert at any time. Given North IS on red alert, the XOR requires South to NOT be on red alert (if both were, the "never both" clause is violated). Option A ignores the exclusivity. Option C would apply to inclusive OR, but the problem explicitly states "never both." Option D mis-reads "exactly one" as unknown. Option E denies a consistent rule. Note: Test 05 Q26 derived ONE side TRUE from a KNOWN FALSE; Test 06 Q26 derives ONE side FALSE from a KNOWN TRUE. Different direction, different answer letter.

Q27: C. 20%

Chain the conditional probabilities: `P(on_duty AND tactical) = P(tactical | on_duty) × P(on_duty) = 0.25 × 0.80 = 0.20 = 20%`. Option B (15%) uses a common arithmetic slip. Option D (25%) is P(tactical | on_duty) alone and forgets to multiply by the duty rate.

Q28: A. Nolan was briefed this week AND signed the attendance log

The premises chain forward by modus ponens: Level A implies briefed weekly; briefed weekly implies signs log. Nolan is Level A, so applying the chain twice: Nolan is briefed weekly, and therefore Nolan signs the log. Option B stops halfway. Option C inverts the chain. Option D would be correct if the premise direction were reversed (as in Test 05 Q28). Option E makes an unsupported exclusivity claim. Note: Test 05 Q28 was an affirming-the-consequent trap (answer: cannot determine); Test 06 Q28 is a VALID forward syllogism (answer: full chain applies). Recognize the direction before selecting.

Q29: C. 45 minutes

Total = `X1 + X2 + X3 + X4 + X5 = 150`. Substitute known values: `X1 + 15 + 30 + 60 = 150`. Solve: `X1 = 150 - 105 = 45` minutes. The individual split of X4 and X5 does not matter; only their sum is needed. Option E is the trap for candidates who think they need more information.

Q30: C. S1 = false, S2 = true, S3 = true

Test the assignment S1=F, S2=T, S3=T:

S3 is T and says "S1 is false." S1=F, so the statement is true and S3 being T is consistent.

S2 is T and says "S3 is true." S3=T, statement true, consistent.

S1 is F and says "exactly one of S2 or S3 is true." S2=T, S3=T, so both are true, not exactly one. The statement is false; S1 being F is consistent.

All three statements land correctly under this assignment. Test the other options and they each break at least one statement. Option A: S2=T says "S3 is true," but S3=F, contradiction. Option B: S2=F says "S3 is true," S3=T, statement true, but S2=F, contradiction. Option D: S2=F says "S3 is true," S3=T, statement true, but S2=F, contradiction.

Scoring: Test 06

27+	Exam-ready. Your reasoning holds under maximum difficulty. One more full-length under timed conditions, then book Pearson VUE.
23-26	Solid pass expected on the real exam. Drill your two weakest question types for one more week. If you missed Q24 or Q30, review truth-teller and self-reference patterns until they are automatic.
18-22	Borderline. Two more weeks of targeted drill. If your logic (Section C) is weaker than observation (Section B), spend time on Test 04 LBR drill. If observation is weaker, rebuild Scenes 9, 10, 11, 12 from memory three times each.
<18	Stop. Return to Tests 01 through 03 and rebuild the foundation. Do not book a Pearson VUE slot at this score.

What Test 06 adds over Test 05: Different figural patterns (dice, triangle counting, 180 rotation of an L, water reflection) that would appear on the same exam but test different neural pathways. Scene 11 is a motorcade arrival, which is the most common USSS operational context; the license plate, country name, and agent count are the kind of details you would be expected to retain from a real operational briefing. Q30 uses a different self-referential construction than Test 05 Q30; recognize that the unique consistent assignment is (F, T, T) rather than (T, T, F). Passing Test 06 at 23+, after Test 05 at 23+, means you have cleared the highest difficulty in this series.